# A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, II
# Revision Log: Draft 8 to Draft 9 (exposition only)

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## Scope

The second of the four exposition passes requested by David Joyner (Part I: Drafts 9–10). **No mathematical change**: no statement, proof, hypothesis or status label was added, removed or altered. A `diff` of Drafts 8 and 9 shows changes only in the title line, in section and subsection lead-ins, in the pointers of §§1, 1.5, 1.6 and 2.3, in the old §8 (Literature), which was replaced, and in the entry [P1].

**Numbering.** Other Parts cite Part II by number (Part I and the glossary cite Definitions 2.1 and 6.4, Remark 5.6, Corollary 6.11, §6.3 and §7(iii)); none cites §8. The Summary of results is therefore an unnumbered section after §1, and the old §8 is replaced by §8 (Notation and conventions) and §9 (Related work), in the same order as Part I. Comparing the `.aux` label tables of Drafts 8 and 9: every number in §§1–7, including theorems and equations, is unchanged.

## Changes

1. **(a) Glossary.** New §8: a lead-in, `\input{../glossary}` (the shared file, unchanged), and §8.11 "Notation specific to Part II" (seventeen entries with pointers: local notation, $\Phi^z$, $d(\mathrm{St})$ and $\lambda_v$, (I1)–(I11), $\Omega_M$, $V_M$, $\delta_n$, $B$, $M_0$, $M_1$, $g^{1/n}$, $\sigma$, $\mathcal{T}$, Shalika germs $\Gamma_{\mathcal{O}}$, (L1), (L2), $f_{\mathrm{St}}$, $e_D$, $\tilde{f}^{\mathrm{St}}$, $\tilde{f}_{\mathrm{reg}}$, (Ell), (Non), (G1), (G2), $u$-admissible pairs, $(\mathrm{H}'_D)$, $(\mathrm{H}')$). Part II must now be compiled from `part-II/`.

2. **(b) Summary of results.** New unnumbered section after §1, under one page: ten items with labels and pointers (compatible measures; Theorem D; germs at central good elements; orbital integrals of arbitrary functions; linear inputs and the Steinberg-type pair; Theorem E; the required germ; Theorem C$'$ and its converse; failure of $(\mathrm{H}'_D)$; what is left), and one line on the running example. All labels are those already in the paper.

3. **(c) Related work.** The old §8 ("Literature … It is preliminary") is replaced by §9 "Related work": it refers to the review of the whole series in Part I, §11, and adds only what is specific to Part II, in two paragraphs: the inputs on covers (germ expansions [KP2], [FK, §8]; Zou; W.-W. Li) and on the linear groups (Shalika germs, Howe, Harish-Chandra, Rogawski, van Dijk, Kivinen–Tsai, pseudo-coefficients after Kazhdan, the linear Jacquet–Langlands inputs, Plancherel formula and formal degrees, Kottwitz's measures, Zelevinsky's classification); and the role of the

section $\sigma$. The three sentences of the old §8 are kept. **No references were added**; every work mentioned was already in the bibliography. The roles assigned to them were checked against their uses in the body (for example [Zel80] for the classification of tempered representations, [Wal03] for the Plancherel formula and Harish-Chandra's finiteness theorem, [KT, Proposition 2.39] for parabolic descent). [BZ], [Ber] and [vD72] are in the bibliography but not cited in the body; the new text mentions them as background only. A first version cited [KP1] for splitting over small subgroups; since [KP1] is not in Part II's bibliography, the sentence cites [FK, §2] only.

4. **(d) Lead-ins.** Each of §§1–7 now says what it depends on and what notation it introduces. Subsection lead-ins were extended where notation or a hypothesis used later is introduced: §§2.2, 2.3, 3.1, 3.2, 3.3, 5.1, 5.2, 5.4, 6.1, 6.3. The other subsection lead-ins are unchanged.

5. **Part I references.** References to Part I now name Draft 10, "whose numbering is that of Draft 8" (§1 lead-in, §2.3, [P1]); no cited number changed, since Part I's numbering is unchanged.

6. Title line: Draft 9, November 8, 2026. Abstract, authors and acknowledgments unchanged.

## Points for David

1. **Bibliography items not cited in the body.** [BZ], [Ber], [vD72] appear only in the bibliography (and now in §9 as background). They may have been cited in earlier drafts; you may prefer to drop them.

2. As in Part I, the existing one-sentence subsection lead-ins were kept.

## Files

`part-II/mjl_part2_draft9.tex` and `.pdf`; this log (`.tex`, `.pdf`); `README.md` (Part II draft number, build note). Draft 8 is kept unchanged; `glossary.tex` is unchanged.